Problem Set 8 - Functions

Hello students,

the goal of this problem-set is to start using functions.

As you saw in the lectures, functions allow you to write pieces of code that can be called multiple times from a program, enabling code reuse.

Assignment 8.1 - Operations on tridimensional vectors (optional)

Develop a program containing some functions/procedures allowing you to perform some basic operations on tridimensional vectors.

You must create multiple functions to allow the user to:

- create a tridimensional vector;
- initialize a tridimensional vector with random coordinates x, y, z (it should be possible to provide *min_value* and *max_value* for the random);
- print a vector (see example for formatting);
- sum two vectors, returning a new one;
- multiply a vector and a constant, returning a new vector;
- cross product (vectorial product) between two vectors (https://en.wikipedia.org/wiki/Cross_product), returning a new vector.

In your main you must use the functions you wrote to test they work properly. Call some of them using the positional arguments approach and the remaining with the keyword arguments. An example of a possible output of the main is:

```
v0 = (8, 11, 9)
v1 = (5, -2, 0)

v0 + v1 = (13, 9, 9)

6 * v0 = (48, 66, 54)

v0 x v1 = (18, 45, -71)
```

Hint: A useful function to generate random ints in a range is randint.
You can import the function as follows:

```
from random import randint
```

Assignment 8.2 - BMI (optional)

The BMI, or Body Mass Index, is a measurement to approximate the quantity of fat present in someone's body, based on weight and height.
The formula to calculate it is: $BMI = \frac{weight(kg)}{height^2(m^2)}$

The result can be confronted with the following table:

- Underweight: $<$ 18.5
- Normal weight: 18.5 – 24.9
- Overweight: 2 5– 29.9
- Obesity: $>=$ 30 or greater

For example Mr. Muster weights 80kg and is 1.8m tall, so his BMI will be calculated as follows:

$BMI = \frac{80}{1.8^2} = 24.7$

Referencing the aforementioned list, it is considered a normal body weight.

A recent survey performed by the Sanity Union Professional Safety International association has generated a huge dataset of data, regarding users. Your job as a professional data-scientist is to extrapolate meaningful data.

Your boss Reginald Osborn Bright needs the following functions to elaborate the dataset:

- get the BMI for each person: given the people list, return a new list also containing the BMI
- get the person with the highest BMI: given the people list, return a new dictionary representing the person and adding the BMI
- get the person with the lowest BMI: given the people list, return a new dictionary representing the person and adding the BMI
- calculate the average BMI: given the people list, return a float containing the average BMI
- calculate the average BMI per males: given the people list, return a float containing the average BMI
- calculate the average BMI per females: given the people list, return a float containing the average BMI
- present in a pretty way an entry

All calculated BMIs must be rounded to 2 decimals

Write a main to show your functions work properly. Here an example of output:

```
Fist 5 average BMIs:
name: Sarah Taylor height: 1.88 weight: 52 gender: female bmi: 14.71
name: Daniel Mitchell height: 1.53 weight: 45 gender: male bmi: 19.22
name: Jesse Rodriguez height: 1.55 weight: 57 gender: female bmi: 23.73
name: Lisa Moore height: 1.64 weight: 95 gender: female bmi: 35.32
name: Robert Martinez height: 1.64 weight: 72 gender: male bmi: 26.77

Person with highest BMI
name: Jennifer Wright height: 1.62 weight: 107 gender: female bmi: 40.77

Person with lowest BMI
name: Michael Hernandez height: 1.92 weight: 47 gender: male bmi: 12.75

Average BMI: 25.16
Average male BMI: 23.84
Average female BMI: 26.48
```

You are provided an extract from the dataset to develop the functions:

```
people = [
    {"name": "Sarah Taylor", "height": 1.88, "weight": 52, "gender": "female"},
    {"name": "Daniel Mitchell", "height": 1.53, "weight": 45, "gender": "male"},
    {"name": "Jesse Rodriguez", "height": 1.55, "weight": 57, "gender": "female"},
    {"name": "Lisa Moore", "height": 1.64, "weight": 95, "gender": "female"},
    {"name": "Robert Martinez", "height": 1.64, "weight": 72, "gender": "male"},
    {"name": "Patricia White", "height": 1.79, "weight": 80, "gender": "female"},
    {"name": "Carol Thomas", "height": 1.68, "weight": 96, "gender": "female"},
    {"name": "Andrew Hill", "height": 1.77, "weight": 69, "gender": "male"},
    {"name": "Carol Allen", "height": 1.76, "weight": 85, "gender": "female"},
    {"name": "Edward Mitchell", "height": 1.77, "weight": 59, "gender": "male"},
    {"name": "Andrew Robinson", "height": 1.69, "weight": 102, "gender": "male"},
    {"name": "Edward Robinson", "height": 1.64, "weight": 57, "gender": "male"},
    {"name": "Andrew Nelson", "height": 1.8, "weight": 102, "gender": "male"},
    {"name": "James Robinson", "height": 1.75, "weight": 67, "gender": "male"},
    {"name": "Ashley Nelson", "height": 1.7, "weight": 67, "gender": "female"},
    {"name": "Brian Roberts", "height": 1.82, "weight": 101, "gender": "male"},
    {"name": "Jennifer Wright", "height": 1.62, "weight": 107, "gender": "female"},
    {"name": "George Lee", "height": 1.76, "weight": 79, "gender": "male"},
    {"name": "Donna Perez", "height": 1.79, "weight": 73, "gender": "female"},
    {"name": "Emily Brown", "height": 1.92, "weight": 56, "gender": "female"},
    {"name": "Michael Hernandez", "height": 1.92, "weight": 47, "gender": "male"},
    {"name": "George Martin", "height": 1.78, "weight": 67, "gender": "male"},
    {"name": "Jessica Mitchell", "height": 1.76, "weight": 76, "gender": "female"},
    {"name": "Sarah Lewis", "height": 1.55, "weight": 88, "gender": "female"},
    {"name": "Sandra Clark", "height": 1.51, "weight": 82, "gender": "female"},
    {"name": "Brian Rivera", "height": 1.7, "weight": 66, "gender": "male"},
    {"name": "Sandra Williams", "height": 1.8, "weight": 71, "gender": "female"},
    {"name": "Mary Roberts", "height": 1.91, "weight": 73, "gender": "female"},
    {"name": "Mary Harris", "height": 1.62, "weight": 72, "gender": "female"},
    {"name": "Edward Davis", "height": 1.53, "weight": 89, "gender": "male"},
    {"name": "Jessica Williams", "height": 1.73, "weight": 71, "gender": "female"},
    {"name": "Brian Perez", "height": 1.86, "weight": 70, "gender": "male"},
    {"name": "Matthew Scott", "height": 1.64, "weight": 72, "gender": "male"},
    {"name": "Robert Lopez", "height": 1.66, "weight": 47, "gender": "male"},
    {"name": "John Flores", "height": 1.72, "weight": 60, "gender": "male"},
    {"name": "Patricia Adams", "height": 1.78, "weight": 71, "gender": "female"},
    {"name": "Emily Jones", "height": 1.58, "weight": 85, "gender": "female"},
    {"name": "James King", "height": 1.91, "weight": 59, "gender": "male"},
    {"name": "Margaret Allen", "height": 1.68, "weight": 59, "gender": "female"},
    {"name": "George Nguyen", "height": 1.79, "weight": 91, "gender": "male"},
]
```

Assignment 8.3 - Prime numbers (optional)

We want to develop a function that tells us if a number is prime or not.
A prime number is a natural number greater than 1 that is only divisible by 1 and by itself.

On the following wikipedia page some more details about prime numbers:
https://en.wikipedia.org/wiki/Prime_number

Develop a function that, depending on the number given as an input, returns a boolean value specifying if it is prime or not.

Use then the implemented function in your main to discover and print on the screen all the numbers that are prime between 1 and 100.

Assignment 8.4 - Cuboids (optional)

Write a function to calculate the surface of a rectangular cuboid (parallelepiped), given its width, height and depth as input.

Write then another function, with the same inputs, that is able to calculate its volume.

After that, write a program that asks the user to input the sizes of 3 cuboids. A size is valid only if all the measures are greater than 0. Each time that a valid cuboid size (width, height and depth) is inserted, the program must store the volume and area of that cuboid in two separate lists.

The program must then execute a procedure, that calculates and prints on the screen the total surface and total volume of the 3 cuboids.

Assignment 8.5 - Trigonometric series (optional)

Write two functions that are able to calculate the value of *sin(x)* and *cos(x)*, using the following trigonometric series:

$$sin\left(x\right) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots \qquad cos\left(x\right) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$

The two functions must have the following interface (`x` must be in radians):

- `sin(x, num_terms)`
- `cos(x, num_terms)`

Compare the values calculated using the trigonometric series with those calculated with the built-in functions `sin()` and `cos()` of the `math` module. Use a growing number of terms and values of `x` equals to 0, $\pi/6$, $\pi/4$, $\pi/3$ e $\pi/2$.

Hint: in order to avoid large numbers as denominator, which could cause overflow errors, calculate the terms of the trigonometric serie as follows:

$$\frac{x^n}{n!} = \left(\frac{x}{n}\right)\left(\frac{x}{n-1}\right)\cdots\left(\frac{x}{1}\right)$$

Assignment 8.6 - Rock paper scissors (optional)

Develop a software that makes you play "Rock Paper Scissors" against the PC.

The user must first specify the number of games to be played.
For each game the user must then input its entry, which must be validated, and the PC generates its move randomly.
In case it is a tie, the game is not considered (not counted).
After all the games have been played, the program must show the statistics and the percentage of winning for each player.

Use functions to structure your code as well as possible.

Example output:

```
How many matches do you want to play? 3

Game 0
---
Insert your entry: lizard
Invalid entry, please insert a valid entry: scissors
Tie! Both players played 'scissors'
Insert your entry: rock
pc plays: scissors
USER WINS!

Game 1
---
Insert your entry: paper
Tie! Both players played 'paper'
Insert your entry: rock
pc plays: paper
PC WINS!

Game 2
---
Insert your entry: paper
pc plays: rock
USER WINS!


---
The overall winner is: USER!
User won 2 time(s), on average 66.67%
PC won 1 time(s): , on average 33.33%
```